Caputo Dirichlet Heat Flow: Sharp Smoothing, Schatten, and Resolvent Limits

Route-A source-local certificate HCS-C277

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Abstract

We solve the Caputo time-fractional heat equation on $(0,\pi)$ in its exact Dirichlet sine basis and prove inverse-stable subordination, positivity, and contraction for every $0<\beta\le 1$. For $0<\beta<1$ the solution family is not a semigroup; at every positive time it gains exactly two spatial derivatives and belongs to $\mathcal{S}_p$ exactly when $p>1/2$. No source operator ideal is promoted to target arithmetic data.

1 The frozen memory owner

Let $X=L^2(0,\pi)$ and

$$A=-\frac{d^2}{dx^2},\qquad D(A)=H^2(0,\pi)\cap H_0^1(0,\pi).$$

The normalized eigenvectors $e_n=(2/\pi)^{1/2}\sin(nx)$ satisfy $Ae_n=n^2e_n$. We freeze

$$^{C}D_t^{\beta}u(t)+Au(t)=0,\qquad u(0)=u_0\in X,\qquad 0<\beta\le 1. \tag{1}$$

For $0<\beta<1$,

$$^{C}D_t^{\beta}f(t)=\frac{1}{\Gamma(1-\beta)}\int_0^t(t-s)^{-\beta}f'(s)\,ds,$$

and $\beta=1$ means the ordinary derivative. Bounded-domain fractional diffusion-wave problems admit eigenfunction constructions of this type [2]; here the interval permits sharp operator thresholds.

Writing

$$E_\beta(z)=\sum_{k=0}^{\infty}\frac{z^k}{\Gamma(\beta k+1)},$$

the coefficient identity

$$^{C}D_t^{\beta}\frac{t^{\beta k}}{\Gamma(\beta k+1)}=\frac{t^{\beta(k-1)}}{\Gamma(\beta(k-1)+1)},\quad k\ge 1,$$

gives the unique solution family

$$S_\beta(t)u_0=\sum_{n\ge 1}E_\beta(-n^2t^\beta)\langle u_0,e_n\rangle e_n. \tag{2}$$

Pollard proved that $E_\beta(-x)$ is completely monotone on $x\ge 0$ for $0<\beta<1$ [1]. More precisely, if $\eta_\beta(t,s)$ is the inverse-stable density characterized by

$$\int_0^\infty e^{-zt}\eta_\beta(t,s)\,dt=z^{\beta-1}e^{-sz^\beta},$$

then scalar Laplace inversion in (2) yields

$$S_\beta(t)=\int_0^\infty e^{-sA}\eta_\beta(t,s)\,ds. \tag{3}$$

Thus $S_\beta(t)$ is self-adjoint and positive, preserves nonnegative functions, and is contractive. Dominated convergence in (2) gives strong continuity at zero.

2 The sharp memory theorem

We use the spectral Sobolev scale $X^s = D(A^{s/2})$. In particular, $X^2 = D(A) = H^2 \cap H_0^1$; "two derivatives" includes this endpoint.

Theorem 1 (category, smoothing, and Schatten boundaries)**.** *Fix $0 < \beta < 1$ and $t > 0$.*

1. *The strongly continuous family S_β is not a semigroup.*
2. *Within the declared domain $\theta \geq 0$, $A^\theta S_\beta(t)$ is bounded on X if and only if $\theta \leq 1$. Equivalently, within $s \geq 0$, $S_\beta(t) : X \to X^s$ is bounded if and only if $s \leq 2$.*
3. *For every $p > 0$, $S_\beta(t) \in \mathcal{S}_p$ exactly when $p > 1/2$. The endpoint $p = 1/2$ fails.*

Proof. If S_β were a semigroup, the continuous first-mode multiplier $E_\beta(-t^\beta)$ would be an exponential. But

$$E_\beta(-t^\beta) = 1 - \frac{t^\beta}{\Gamma(1+\beta)} + O(t^{2\beta}),$$

which is not exponential when $\beta < 1$.

On the positive ray, the standard Mittag–Leffler expansion gives

$$E_\beta(-x) = \frac{1}{x\Gamma(1-\beta)} + O(x^{-2}) \quad (x \to \infty). \tag{4}$$

If a later reciprocal-Gamma coefficient vanishes, (4) only improves. Hence

$$n^{2\theta} E_\beta(-n^2 t^\beta) \sim \frac{t^{-\beta}}{\Gamma(1-\beta)} n^{2\theta-2}.$$

Continuity on compact x-intervals supplies the upper bound at $0 \leq \theta \leq 1$, and the positive asymptotic proves failure at every $\theta > 1$. If $\theta < 0$, then $A \geq I$ makes A^θ bounded, so $A^\theta S_\beta(t)$ is also bounded; these negative powers lie outside the declared $\theta \geq 0$ smoothing domain. The singular values themselves are asymptotic to a positive constant times n^{-2}, so their pth powers sum exactly for $2p > 1$. $\square$

The endpoint is substantive: memory retains only one bounded power of the second-order generator. This differs from merely asserting some unspecified regularization, and it prevents importing the heat flow's analytic smoothing into the fractional clock.

References

[1] H. Pollard, *The completely monotonic character of the Mittag-Leffler function $E_a(-x)$*, Bull. Amer. Math. Soc. 54 (1948), 1115–1116, doi:10.1090/S0002-9904-1948-09132-7.

[2] K. Sakamoto and M. Yamamoto, *Initial value/boundary value problems for fractional diffusion-wave equations and applications to some inverse problems*, J. Math. Anal. Appl. 382 (2011), 426–447, doi:10.1016/j.jmaa.2011.04.058.